

Mehran Khodabandeh

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Senior ML Scientist and Engineer (+6 years) | PhD in C.S. | Deep Learning Specialist | Startup Founder

With a PhD in Computer Science and +6 years of hands-on experience in deep learning, I've honed my skills at leading tech giants such as Google, Microsoft, and Meta. As an ML engineer, entrepreneur, and founder of my own startup, I am passionate about driving innovation and creating simple and effective solutions.

» Professional Experiences

- Research Engineer at Meta (Superintelligence Lab)**, Bellevue, WA Nov. 2024 - Present
- Drive safety alignment of large language models by leading fine-tuning, data curation, and large-scale evaluation efforts to reduce harmful outputs and improve user experience.
 - Design and implement novel methods and experiments to ensure LLMs comply with safety guidelines.
- Co-founder & Machine Learning Scientist** at Variational.AI, Vancouver, BC Oct. 2019 - Nov. 2024
- At Variational AI we used AI to discover new molecules that will become drugs
 - Led a multidisciplinary team of four people: two ML scientists/engineers, one software engineer, and one computational chemist.
 - Contributed to the creation and optimization of generative models (diffusion models, VAEs, etc.) for the discovery of novel drug molecules.
 - Addressed and resolved multifaceted challenges in data processing, development of deep learning models, and software engineering aspects of projects
- Research Internship at Facebook AI Applied Research**, Menlo Park, CA Summer 2020
- Worked on [Video Instance Segmentation using Transformers](#)
- Software Engineer Internship at Google Research**, Seattle, WA Summer 2018
- Worked on [image segmentation for 360 images](#)
- Research and Development Internship at D-Wave**, Burnaby, Canada Spring 2018
- Focused on [domain adaptation in object detection](#), advancing detector performance in novel environments.
 - Secured a US patent and published a paper at ICCV 2019.
- Research Internship at Microsoft**, Redmond, WA Summer 2017
- Worked on video generation (GANs), addressed training data scarcity for action recognition
 - Acquired a US patent and showcased findings at CVPR Workshop 2018.
- Research Internship at National Institute of Informatics(NII)**, Tokyo, Japan Summer 2016 and 2015
- Worked on [active learning](#) for structured data, implemented in Caffe (Python)
 - Presented the work at ICPR 2016 (oral)

» Education

- **Ph.D at Simon Fraser University**, Burnaby, BC, Canada. Sep 2015 – Oct 2019
Computing Science (Machine Learning) - Supervisor: [Dr. Greg Mori](#)
Took a four-year break to work on my startup. The actual graduation date is August 2023

Mehran Khodabandeh

- **M.Sc. at Simon Fraser University**, Burnaby, BC, Canada. Jan 2013 – Jan 2015
Computing Science (Computer Vision and ML) – GPA: A/A+ - Supervisor: Dr. Greg Mori
- **B.Sc. Sharif University of Technology**, Tehran, Iran. Sept 2008 – June 2012
Bachelor Degree in Computer Engineering (Hardware Engineering) – GPA: 17.45/20

» Selected Publications

- **Strongest Path: A Cytoscape Plugin to Find The Most Confident Paths in Protein-Protein Interactions Networks.** 2021
BMC bioinformatics
(Z. Mousavian, M. Khodabandeh, A. Sharifi Zarchi, A. Nadafian, A. Mahmoudi)
<http://apps.cytoscape.org/apps/strongestpath>, <https://github.com/strpaths/release>
- **A Robust Learning Approach to Domain Adaptive Object Detection¹** 2019
ICCV 2019
(M. Khodabandeh, A. Vahdat, M. Ranjbar, W. G. Macready)
<https://arxiv.org/abs/1904.02361>
- **Human Action Video Generation for Training Activity Recognition Models** 2018
CVPR Workshop 2018
(M. Khodabandeh, Hamidreza Vaezi Joze, Ilya Zarkhov, Vivek Pradeep)
[PDF](#)

» Patents

- **Systems and methods for domain adaptation**, US Patent, 16/779,035 2020
 - M. Khodabandeh, A. Vahdat, M. Ranjbar, WG Macready, Z Bian
- **Human action data set generation in a machine learning system**, US Patent, 10,679,044 2020
 - M. Khodabandeh, HV Joze, I Zharkov, V Pradeep, M Khodabandeh

» Skills

Programming Languages: Python, Java, Javascript, C, HTML, CSS, C++, MySQL. (ordered by proficiency)
Frameworks: PyTorch, TensorFlow

» Honors and Awards

- Received Graduate Fellowship at Simon Fraser University (Canada) 2013 - 2017
- Granted admission for Master of Science from Talented Students Office of Sharif University of Technology (Iran) Fall 2012
- Ranked (248th) amongst 500,000 students in Iranian National University Entrance Exam Fall 2008